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A Probabilistic Approach to Adaptive Protection in the Smart Grid

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Smart grids are critical cyber-physical systems that are vital to our energy future. Smart grids' fault resilience is dependent on the use of advanced protection systems that can reliably adapt to changing grid conditions. The vast amount of operational data generated and collected in smart grids can be used to develop these protection systems. However, given the safety-criticality of protection, the algorithms used to analyze this data must be stable, transparent, and easily interpretable to ensure the reliability of the protection decisions. Additionally, the protection decisions must be fast, selective, simple, and reliable. To address these challenges, this article proposes a data-driven protection strategy, based on Gaussian Discriminant Analysis, for fault detection and isolation. This strategy minimizes the communication requirements for time-inverse relays, facilitates their coordination, and optimizes their settings. The interpretability of the protection decisions is a key focus of this article. The method is demonstrated by showing how it can protect the medium-voltage CIGRE network as it transitions between islanded and grid-connected modes and radial and mesh topologies.

CCS Concepts: • **Hardware** → **Smart grid**; *Safety critical systems*; • **Computer systems organization** → Embedded and cyber-physical systems;

Additional Key Words and Phrases: Adaptive protection, microgrid, Gaussian Discriminant Analysis, machine learning, over-current inverse time

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1 Introduction

The smart grid is a critical cyber-physical system that plays a central role in modernizing the electric power grid. Through leveraging digital technology, communication, and automation, the smart grid stands to enhance energy security, efficiency, and sustainability [14]. A key enabler of these benefits is the smart grid's ability to integrate renewable energy sources and promote grid flexibility through advanced monitoring and control.

Grid flexibility involves adapting and responding to changes in demand, generation, and operating conditions to maintain reliable and efficient performance [14]. A critical aspect of this flexibility is

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the smart grid's capacity to rapidly reconfigure its connectivity, generation, and physical topology in response to shifts in operating conditions or disturbances [7, 14]. These adaptive measures optimize electricity flow, enhance system reliability, and ensure a consistent power supply even in the face of disturbances.

Microgrids serve as a prime illustration of the smart grid's flexibility. Microgrids are small-scale, localized energy systems that seamlessly integrate various distributed energy resources, including renewables [27]. Additionally, microgrids possess the flexibility to operate either in isolation or in connection to the main power grid. The microgrid is said to be islanded when in isolation from the main grid.

Unfortunately, these advancements pose challenges for smart grid protection. Power system protection involves protection devices that detect and isolate faults to minimize damage, maintain system stability, and prevent failure [5]. Protection devices must act rapidly and selectively, isolating faults while enabling the rest of the system to continue operation.

Smart grid protection, particularly at the distribution level within microgrids, proves to be a highly challenging task. Challenges arise from bidirectional power flow originating from distributed energy generators and the highly flexible grid topology [11, 29]. As microgrids transition between different topology configurations and between grid-connected and islanded modes of operation, protection schemes must dynamically adapt to these changing grid conditions [7, 20]. Consequently, enabling the smart grid is dependent on the development of reliable protection schemes capable of adapting to these challenges.

Adaptive protection is regarded as the most suitable approach to protecting microgrids [11, 39]. Adaptive protection entails the design of protection schemes capable of dynamically adjusting their operating characteristics in response to changes in the grid's operating conditions. This adaptability enables the protection system to effectively respond to various fault scenarios under different grid conditions [24, 39].

Adaptive protection is commonly achieved by extending the functionality of existing traditional protection devices, such as inverse-time over-current relays, which constitutes a cost-effectiveness benefit in the absence of specialized commercial microgrid relays [20] and the desire to easily integrate protection schemes into existing devices and substations with minimal capital expenditure [2].

Supported by years of operation in traditional power systems, over-current relays offer an economical and effective solution to protect the power system against faults. With the rise of microgrids, there is a growing interest to extend the capabilities of over-current relays to enable adaptive protection [10, 46, 47]. This requires the development of methods to update over-current relay settings in response to changes in the grid while maintaining both selectivity and sensitivity [18]. Selectivity refers to the relay's ability to precisely locate and classify faults to isolate the smallest possible faulty section of the power system, while sensitivity refers to the ability to promptly and accurately detect faults to quickly clear them.

Recent studies have explored the use of data-driven techniques, including **Machine Learning (ML)** algorithms, to leverage the growing amount of data generated in power systems to satisfy these protection requirements. Data-driven techniques have been applied to detecting faults [28, 29], identifying the location, type, and phase of faults [1, 23], and adjusting relay settings [43]. Based on the findings of these studies, it appears that data-driven techniques can offer several benefits. Firstly, data-driven methods can enhance the accuracy of protection decisions, improving fault detection and classification and reducing unnecessary tripping. For example, Mishra et al. [30] conducted a study where they compared the accuracy of fault detection in protection schemes based on **Decision Trees (DTs)** and **Random Forests (RFs)** with traditional over-current relays. Their study showed over 40% improvement in fault detection accuracy when leveraging the ML

algorithms. Secondly, data-driven techniques can offer a scalable solution to adapting relay settings, significantly reducing the need to manually identify system topology configurations and compute relay settings. For example, in centralized communication-based implementations, data-driven techniques have the potential to replace traditional relays, monitoring the entire power system and directly controlling circuit breakers [39]. We survey some of the literature applying data-driven techniques to adaptive protection in Section 2.

Data-driven techniques, however, come with their own set of limitations and potential threats to protection reliability. Black-box algorithms, such as **Neural Networks (NNs)** and RFs, while often highly accurate, cannot be easily verified or validated, making them a significant risk for adoption in safety-critical applications such as protection. Some other algorithms such as DTs, while more interpretable, may suffer from instability: inconsistent decisions resulting from small changes in input data. This instability can also make it difficult to validate their decisions, thus undermining their feasibility for protection. To successfully integrate data-driven techniques into adaptive protection, it is crucial to ensure that the interpretability of the protection decisions is preserved. This enables protection engineers to verify the decisions while simultaneously reducing potential threats to reliability.

Additionally, adaptive protection schemes, which rely on real-time communication infrastructure such as peer-to-peer communication between relays or communication between a central protection management center and relays, may face potential threats to reliability due to network failure, latency, and cyberattacks [17, 18]. Although increased communication can lead to more accurate and informed protection decisions, the associated risks and threats must be carefully considered.

To acquire the benefits of data-driven techniques and communication-based adaptive protection while reducing threats to reliability, we propose a novel data-driven adaptive protection scheme that emphasizes the interpretability of protection decisions for validation and minimizes reliance on communication. Our proposed approach is based on **Gaussian Discriminant Analysis (GDA)** and is designed to (1) identify the system topology configurations that require a change in the protection relays' fault response, (2) specify necessary communication with relays, (3) locate faults with respect to relays for relay coordination, and (4) optimize relay settings for fast fault clearing. This approach is centralized with a central **Microgrid Protection Management Center (MPMC)** communicating with relays following infrequent major grid changes. To demonstrate the effectiveness of our proposed strategy, we apply it to inverse-time protection relays.

1.1 Contributions

- Our work presents a first-of-its-kind application of GDA to adaptive protection. Through our research, we discuss and demonstrate several benefits of GDA for adaptive protection, including improved data- and computational-efficiency, interpretability for ease of validation, adaptability to changing grid conditions, timely response, and reduced reliance on communication for protection. To showcase the interpretability of our proposed strategy, we put emphasis on visualizing its protection decision-making process in this article.
- We extend the inverse-time protection formula to improve fault distinction by incorporating voltage drops associated with faults.
- We formulate an optimization problem that enables the coordination and optimization of relays in a changing system environment.

We validate the strategy on the **Conseil International des Grands Réseaux Electriques (CIGRE)** medium voltage benchmark system. The benchmark system permits many configurations to help validate the proposed adaptive protection method, including radial and mesh topology, and grid-connected and islanded modes.

1.2 Organization

The rest of the article is structured as follows. We survey related work in Section 2. We present background material in Section 3. We present the problem statement discussing the goals of data-driven adaptive protection in Section 4. We develop the protection strategy in Section 5. Next, we illustrate and discuss the results of the protection strategy using the benchmark system in Section 6. The conclusion is in Section 7.

2 Related Work

The literature reviews in [7, 17, 24, 40] provide different perspectives on adaptive protection. We summarize the contents of these reviews that are relevant to our work. Our work proposes an intelligent centralized protection scheme with pre-set relays, which involves a central MPMC issuing instructions to the relays to switch their settings only after significant topology changes. Next, we survey some of the literature that applies data-driven techniques to adaptive protection to motivate our work.

In their analysis of literature on adaptive protection in the context of communication, Gutierrez-Rojas et al. [17] categorize their examination according to communication technology (wireless or wired) and control approach (centralized or decentralized). Most of the adaptive protection research proposes centralized control approaches. In centralized control, a central MPMC is established to monitor the power system in real-time and send (either wired or wireless) instructions to relays to switch between relay settings or directly to circuit breakers to trip. For example, Oudalov et al. [32] implemented a centralized control approach in which wired communication is established between the control center and advanced circuit breakers with integrated over-current relay units. In real-time, data are collected from the circuit breakers and relay settings are modified. Alternatively, communication can be limited to notifying relays of infrequent major grid topology changes, such as network reconfiguration, as in [10], with local measurements used to detect other changes, such as distributed energy resource status.

Data-driven techniques can also be employed to detect changes. For example, Ozcanli et al. [33] trained one-dimensional **Convolutional Neural Network (CNN)** and **Long Short-Term Memory (LSTM)** models to detect the isolation of microgrids from the main grid. The models analyzed voltage and current harmonics at the **Point of Common Coupling (PCC)** between the microgrid and the main grid.

Conversely, Barra et al. [7] organize their review by methods that rely on computational intelligence, such as ML, fuzzy logic, and optimization, and methods that do not. Khalid et al. [24] categorize their review based on articles that propose methods for fault detection, relay setting adjustment to changes in network configuration, or relay optimization for coordination and articles with methods that require and do not require pre-setting relays.

Recent studies have explored the use of data-driven techniques to make use of the growing streams of data in power systems for more accurate and reliable adaptive protection.

One notable utilization of data-driven techniques in adaptive protection involves the reduction of setting groups for protection relays. Protection relays store in their device memory the operational settings that determine their response to faults. In adaptive protection schemes, relays exhibit adaptability to changes in grid configuration by switching between different relay modes—each with a different group of operational settings. While there could be numerous grid configurations, it is not necessary for a protection device to adjust to each configuration. The process of reducing setting groups aims to identify a significantly smaller number of setting groups for the protection devices. This approach offers advantages such as reduced memory requirements for storing settings

in the protection devices and decreased communication needs to notify the devices of changes in the grid configuration.

The identification of setting groups has been achieved through the application of unsupervised learning techniques. Alam et al. [4] implemented k -means clustering, Barik et al. [6] utilized k -medoids, and Saldarriaga et al. [37] adopted a hybrid approach that combines k -means clustering with Principal Component Analysis for the purpose of efficiently identifying setting groups. Conversely, authors in [34, 38, 42] employed various supervised learning feature selection techniques, including **Recursive Feature Elimination (RFE)** and **meta-transformers (mTs)**.

Other applications of data-driven techniques in adaptive protection involve enhancing the response of protection devices to faults. The survey by Shakiba et al. [40] reviews the use of data-driven techniques in identifying (or detecting) the presence of a fault, classifying the type of fault (phase-to-ground, phase-to-phase, etc.), and locating the fault to isolate it.

Data-driven techniques can be broadly classified into two categories: traditional ML and deep learning methods. We use traditional ML to refer to conventional statistical approaches, while deep learning involves methods utilizing NNs. Below, we first explore studies employing ML, followed by a review of recent studies utilizing deep learning.

Mishra et al. [30] employed DTs for fault detection and classification, achieving 97% accuracy in fault detection. The results were demonstrated using the **The Consortium for Electric Reliability Technology Solutions (CERTS)** microgrid model [26]. Expanding on the use of DTs, Saleh et al. [38] constructed a more complex algorithm leveraging a DT ensemble to identify fault zones and another DT ensemble for fault classification.

Pre-processing the input data to DTs to extract statistical features through wavelet transforms applied to current measurements, Tang et al. classified faults locally by employing a DT at each relay. They proposed a peer-to-peer communication system between relay devices instead of relying on communication from the central MPMC. An additional novelty in Tang et al., beyond fault detection, classification, and localization, was the use of an **Artificial Neural Network (ANN)** to adjust relay settings.

Abdelgayed et al. [1] applied various algorithms to statistical features extracted through wavelet transforms for fault type classification. Notably, their **k-Nearest Neighbors (kNN)** achieved 95.63% accuracy, DT 90.4%, **Support Vector Machine (SVM)** 93.3%, and naïve Bayes 94.24%. Their results were demonstrated using the CERTS microgrid model.

SVMs have recently demonstrated increased application and high performance in data-driven adaptive protection. Aiswarya et al. [3] proposed a dual-SVM adaptive protection scheme, with the first SVM responsible for identifying system conditions and detecting faults and the second SVM responsible for fault classification and localization. Srivastava et al. [41] utilized SVM and Gaussian Process Regression for fault detection and localization, respectively. Their methods outperformed DTs and bagged trees in fault detection and outperformed support vector regression and DT ensembles in fault localization. Moradzadeh et al. [31] further corroborated SVMs' superior performance in fault classification and localization compared to other ML methods, including DTs and kNN. Lin et al. [29] achieved fault localization accuracy exceeding 99% using SVMs investigating the protection of the IEEE 9 bus system.

Moving to review the use of deep learning in adaptive protection, Lin et al. [28] trained two ANN models for fault classification and localization based on current measurements. Similarly, Zayandehroodi et al. [48] used combinations of radial-basis function ANNs for fault localization. Thomas et al. [44] proposed a hybrid CNN-Transformer model for fault detection, classification, and localization.

Contrasting the use of raw current measurements, studies like those of James et al. [23], Kim et al. [25], and Rizeakos et al. [35] trained algorithms on statistical features extracted from wavelet

transforms. James et al. [23] employed deep recurrent ANNs and achieved fault detection, fault-type, and -phase classification accuracies exceeding 97%, with over 94% accuracy in fault localization. Their study was demonstrated using the CERTS microgrid and IEEE 34-bus systems. Rizeakos et al. [35] used a CNN model for fault localization and classification, while Kim et al. [25] trained an LSTM model for identifying fault direction and coordinating protection.

Moradzadeh et al. [31] explored various deep learning methods, including CNN, LSTM, and a novel hybrid CNN-LSTM model. Their comparison against ML methods, including SVM, DT, and kNN, revealed CNN outperforming both ML and LSTM in fault classification and localization. The hybrid CNN-LSTM model achieved the highest performance.

While deep learning exhibits high accuracy and proves valuable in addressing the complexities of adaptive protection, its adoption for adaptive protection faces significant challenges due to its black-box nature and the inherent difficulties in verification and validation. Several traditional ML methods, though potentially outperformed by deep learning in terms of accuracy, compensate with greater interpretability, rendering them more suitable for safety-critical applications like adaptive protection. Among traditional ML methods, a nuanced comparison reveals specific advantages for specific methods over others. Despite the interpretability of DT, they suffer from instability, where minor changes in the training data can lead to substantial alterations in the DT's structure, resulting in significantly different hypotheses and undermining interpretability [13]. The kNN algorithm's drawback lies in its slowness during test time, involving the calculation of the distance between the test instance and all training data. As the size and dimensionality of the training data increase, the computation time for making predictions also increases. This drawback makes kNN impractical for real-time applications with large datasets that require timeliness. Naïve Bayes and GDA are both probabilistic data-driven techniques with similarities. GDA stands out for its ability to learn more complex features, potentially achieving higher accuracy. It is worth noting that probabilistic data-driven methods, in general, offer stability, intuitiveness, interpretability, and computational and data efficiency [15, 16]. We show that SVM and GDA are comparable in performance; the rationale for choosing GDA over SVM is discussed in Section 6.

3 Background

We apply our adaptive protection method to the optimization and coordination of inverse-time protection relays. We begin this section by describing the operation of time-inverse relays. Next, we introduce GDA, which is the probabilistic data-driven approach that we use to distill fault data into protection decisions. Lastly, we introduce **Conditional Value-at-Risk (CVaR)**, which we use in parametrizing the fault data for relay optimization.

3.1 Inverse-Time Protection

Over-current relays apply the following inverse-time equation [9] to compute their operation time: the delay before the relay trips

$$t_{op}(I_M; \text{TMS}, \eta, \lambda, L, I_S) = \text{TMS} \left(\frac{\lambda}{\left(\frac{I_M}{I_S}\right)^\eta - 1} + L \right), \quad (1)$$

where $t_{op} > 0$ is the relay operation time, I_M is the relay current measurement, and **Time Multiplier Setting (TMS)** and I_S are the time-multiplier and pick-up current settings of the relay, respectively. Directional over-current relays measure current in one direction making them useful for protecting mesh and multi-source grids.

Equation (1) defines a curve that is shaped by characteristic constants λ , η , and L . Adaptive protection involves shifting the curve to cope with changing grid conditions [12].

Consequently, adaptive protection identifies distinct modes of operation for the relay, each associated with a unique setting group $\{\text{TMS}, \eta, \lambda, L, I_S\}$. This allows the relay to operate differently in each mode, adapting to different grid conditions.

Inverse-time over-current relays work on the principle that fault currents decrease with increasing distance to the fault location. As such, shorter operation times are computed per Equation (1) for nearer faults. This facilitates relay coordination where relays closer to the fault location trip earlier to isolate the smallest possible section of the power system.

Anticipating the challenge of coordinating relays only based on fault currents in distribution systems, such as microgrids, due to limited fault currents, we extend Equation (1) to consider fault voltage

$$t_{op}(I_M, V_M; \text{TMS}, \eta_i, \eta_v, \lambda_i, \lambda_v, L_i, L_v, I_S, V_S) = \text{TMS} \left(\frac{\lambda_i}{\left(\frac{I_M}{I_S}\right)^{\eta_i} - 1} + L_i \right) \left(\frac{\lambda_v}{\left(\frac{V_S}{V_M}\right)^{\eta_v} - 1} + L_v \right), \quad (2)$$

where V_M is the relay voltage measurement and V_S is the pick-up voltage settings of the relay. Setting $\lambda_v = 0$ and $L_v = 1$ in Equation (2) yields Equation (1).

In Equation (2), the trip time is inversely proportional to the current magnitude and inversely proportional to the voltage magnitude. This means that the protective device trips faster for larger currents and voltage drops and slower for smaller currents and voltage drops.

3.2 GDA

GDA is a probabilistic data classification algorithm that fits a Gaussian distribution to each class of data. Let $y \in \{1, 2, \dots, n\}$ represent the classes, and n be the number of classes. The probability of sampling an observation (or measurement) $\mathbf{x}$ given class $y = k$ is

$$P(\mathbf{x}|y = k) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k), \quad (3)$$

where $\boldsymbol{\mu}_k$ and Σ_k are the mean vector and covariance matrix of the Gaussian distribution fitted to class k , respectively. A GDA classifier infers the observation's class via Bayes' theorem:

$$P(y = k|\mathbf{x}) = \frac{P(\mathbf{x}|y = k)P(y = k)}{\sum_{i=1}^n P(\mathbf{x}|y = i)P(y = i)}. \quad (4)$$

GDA can also be used for supervised dimensionality reduction to project the data to a reduced subspace, which maximizes the separation between classes. Following an assumption that all classes share the same covariance matrix of $\Sigma_y = \Sigma$ for $y = \{1, 2, \dots, n\}$, we can choose this reduced subspace to be linear, which aids in interpreting the results of GDA. The resulting algorithm is termed **Linear Discriminant Analysis (LDA)**. Taking the log-posterior of Equation (4) gives the following equation:

$$\log P(y = k|\mathbf{x}) = (\Sigma^{-1} \boldsymbol{\mu}_k) \mathbf{x} + C \quad (5)$$

where the coefficients in $\Sigma^{-1} \boldsymbol{\mu}_k$ determine the importance of the observations' features in separating the classes (C includes other unimportant terms). Below a certain threshold, unimportant features can be neglected to reduce the dimensionality of the problem.

3.3 Conditional Value-At-Risk

CVaR [36] is a risk measure of the expected value of a random variable in the lowest $100\alpha\%$ of cases. Let X be the random variable and q_α be the 100α percentile value of the distribution, then

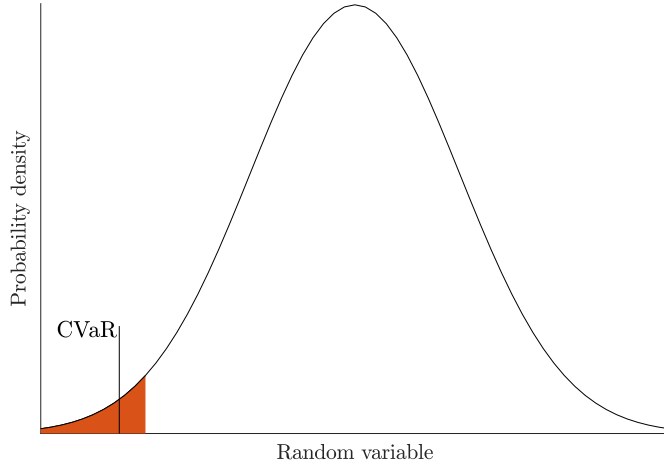


Fig. 1. Illustrating the CVaR of a Gaussian distribution. The red area under the curve represents the $100\alpha\%$ tail of the Gaussian distribution. The CVaR_α value is the mean of this area and is represented by the vertical line.

the CVaR with a confidence interval of $100\alpha\%$ is

$$\text{CVaR}_\alpha = E(X|X \leq q_\alpha). \quad (6)$$

For a Gaussian distribution, CVaR has the following closed form:

$$\text{CVaR}_\alpha = \mu - \sigma \frac{\phi(\Phi^{-1}(\alpha))}{\alpha} \quad (7)$$

where μ and σ are the distribution's mean and standard deviation, respectively; ϕ is the density function; and Φ^{-1} is the standard normal quantile. CVaR_1 is equivalent to the mean of the random variable.

Figure 1 demonstrates the area representing the $100\alpha\%$ tail of the normal distribution in red. CVaR_α is the mean of this area.

4 Problem Statement

We use the CIGRE medium voltage distribution network, illustrated in Figure 2, as the benchmark system in this research. This section introduces the network in advance of the experiments section to elucidate the challenges addressed in the research.

The CIGRE network offers an ideal model to illustrate the challenges associated with adaptive protection and to showcase the performance of adaptive protection methods. The system consists of two radial microgrids (Microgrid 1 and Microgrid 2 in Figure 2) but can switch to a complex mesh topology by closing switches S1, S2, and S3. The microgrids can operate in islanded mode where each microgrid is isolated with **Distributed Generators (DGs)** DG2 and DG3 supplying entire loads of their respective microgrid; islanded interconnected microgrid mode where both microgrids are connected but not to the main grid; or grid-connected mode. Consequently, the benchmark system allows for multiple configurations, facilitating the validation of adaptive protection methods.

The PCCs for the microgrids (PC1 and PC2) can be open or closed. For instance, closing PC2 connects Microgrid 2 to the main grid. Additionally, each DG can be online or out-of-operation.

The solid thick horizontal nodes represent system buses, each numbered. Power lines connecting the buses have two directional over-current relays responding to faults in opposite directions

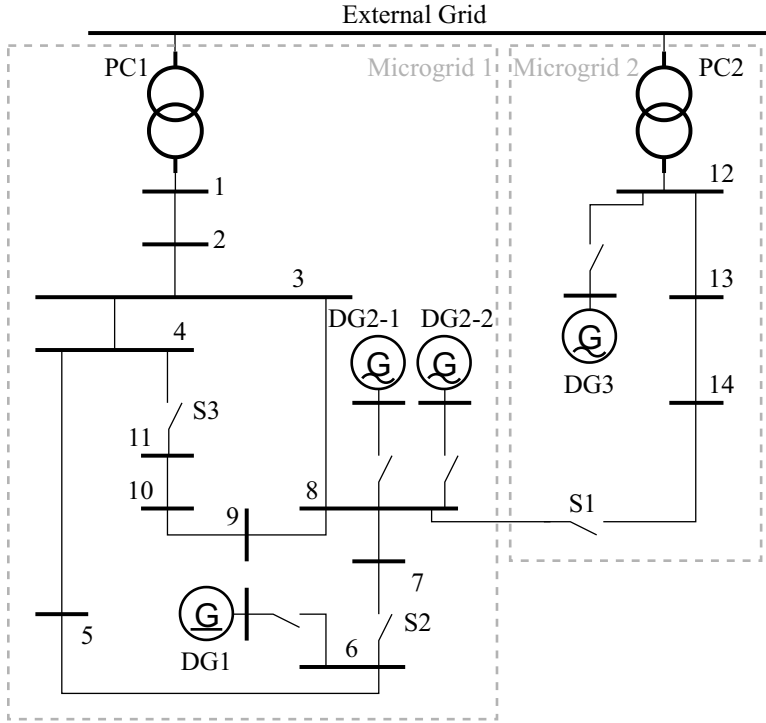


Fig. 2. CIGRE medium voltage distribution network [8, 20]. It consists of two microgrids. S: switch; PC: point-of-common coupling; DG: a distributed generator.

along the power line. For instance, the power line between buses 12 and 13 in Microgrid 2 has a directional over-current relay, R12-13, responding to downstream fault currents traveling from bus 12 to 13. Similarly, a directional relay, R13-12, near bus 13, responds to fault currents traveling from bus 13 to 12.

When a power system experiences a short-circuit fault, the sudden reduction in impedance due to the introduction of the short-circuit causes a current rush. Over-current relays must measure the increased current and isolate the fault to prevent significant consequences, including power system instability or equipment damage. Primary relays, closest to the fault, must trip earliest to isolate the smallest possible section of the power system, with backup relays coordinated to trip in case the primary fails. In practice, primary relays should operate following a short delay (or dead time), D_{min} , to allow temporary faults to clear on their own, and backup relays should operate following a preset **Minimum Coordination Time (MCT)** delay to coordinate with the downstream relays in case of relay malfunction.

The flexibility of the CIGRE network poses several challenges:

- (1) Fault currents differ significantly between grid-connected and islanded modes. For instance, relay R12-13 observes markedly different fault currents for a downstream fault at bus 14 depending on the open or closed status of PC2. When PC2 is closed, the short-circuit current capacity of the main grid significantly surpasses that of DG3. Given that the operation time of over-current relays is contingent on the current magnitude, the absence of relay

setting adjustments between grid-connected and islanded modes can lead to a considerably delayed response to faults in islanded mode due to the lower short-circuit current capacity.

- (2) Fault currents differ due to network topology changes. Relay R12-13, for instance, observes different fault currents for a fault at bus 14 depending on the state of switch S1. When S1 is closed, fault currents from Microgrid 1 contribute to the short circuit at bus 14, altering the fault current pulled through the line between buses 12 and 13 that relay R12-13 monitors. Additionally, the closing of S1 requires relay R12-13 to coordinate with Microgrid 1's relays.
- (3) Not all grid configurations necessitate unique relay settings. Within the CIGRE network, there exist 2^9 distinct configurations stemming from the on/off statuses of the two PCCs, three switches, and four generator units. The exponential growth in the number of configurations with the grid size poses operational challenges, imposing substantial communication burdens on the MPMC and relays. Moreover, it burdens the memory of relays, which must store settings for all configurations in memory. Consequently, it is imperative to discern, for each relay, a minimal set of settings that ensures effective grid protection while mitigating operational complexities and resource burdens.
- (4) Mesh topology demands meticulous relay coordination. A fault at bus 8, with all switches closed, requires the operation of multiple primary relays (R3-8, R9-8, R7-8, and R14-8). Moreover, backup relays must be on standby in case the primary relays fail. For example, for a fault at bus 8, relays R4-3 and R2-3 serve as backups to relay R3-8. Backup relays wait for an MCT to confirm the primary relay's failure before operating, ensuring selective fault isolation. Failure to coordinate between primary and backup relays results in relay in-coordination, either leading to defective protection systems that fail to isolate faults or needlessly isolating large grid sections. Mesh topology imposes complex coordination requirements on numerous relays to effectively isolate faults.

Our data-driven adaptive protection method distills data into protection decisions that address the following goals: (1) fault identification and classification, (2) relay optimization and coordination, and (3) communication specification. We discuss these goals below.

In relation to a relay, faults can be classified according to their distance from the relay. A fault is primary (to a relay) if the relay is the closest device to the fault location. The next (closest) relay(s) are backup to the primary relay. A fault can have multiple primary relays and a relay can have multiple backups in situations of equidistant relays from the fault. In this article, we use the terms secondary and tertiary faults to classify faults that are progressively further from a relay, and other to collectively classify tertiary and further faults. *Fault identification and classification* involve enabling relays to accurately identify the existence and class of a fault from local current and voltage measurements. Identifying the class of the fault helps relays locate the fault, which enables their coordination. *Relay optimization and coordination* involve computing relay settings to ensure rapid, correct, selective, and coordinated protection.

Variations in power system operating conditions can significantly change the currents measured by relays, which can cause relay misoperation. Relays must adjust their operation in response to major grid changes, including the transition between grid-connected and islanded modes of operation, grid topology, and generator availability, which can significantly affect fault current. This involves communicating (typically per the IEC61850 [21] protocol) these grid changes to the relays and directing them to switch their relay settings. It is desirable to minimize the reliance on communication to reduce the infrastructure capital costs and system complexity and prevent risks associated with communication delays and cyberattacks. *Relay communication specification* involves specifying the contents of the information that should be communicated to the relays.

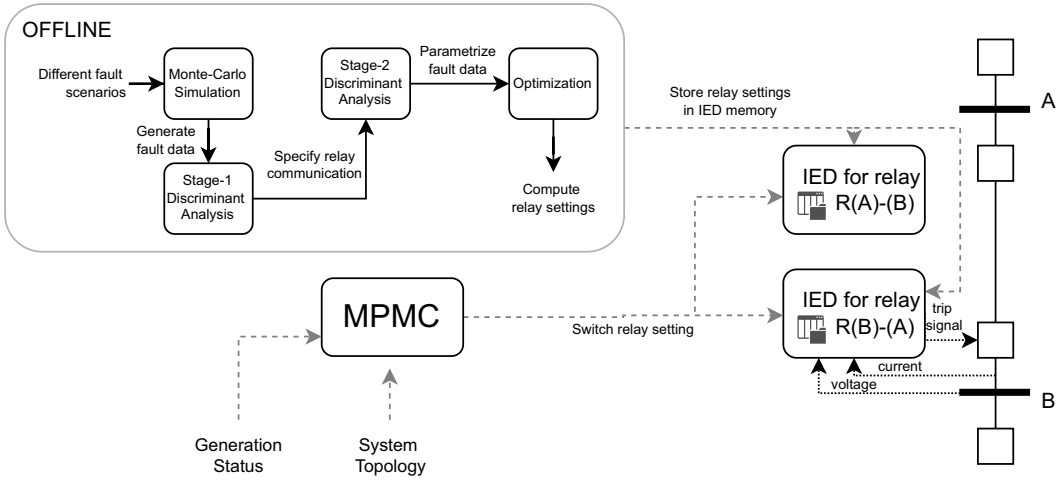


Fig. 3. Flowchart of the proposed method.

By limiting communication with relays to actionable information that signals relays to switch into a mode of operation better suited to current grid conditions, relay communication specification also involves specifying the relays' modes of operation.

Finally, a comprehensive protection strategy must consider immeasurable, uncertain factors, including fault impedance, renewable energy intermittency, and fault distance from relays, which also influence the fault current.

The next section introduces a GDA-based method that addresses these goals.

5 Proposed Adaptive Protection Method

Terminology. We introduce the following terminology before presenting the method: We use $P(X|Y; \theta)$ to denote the probability of a random variable X given Y . The distribution parameters are represented by θ . We denote directional relays with $RA-B$, where A is the bus it is located at and B is the bus it is directed to. Each relay stores multiple relay setting groups. A setting group specifies the TMS, characteristic constants, and pick-up settings used by the relay to compute its operation time. Each group specifies a distinct relay mode of operation, which is dependent on the state of the system. The superscript on variable $X^{(r)}$ associates it with a specific relay r and the uppercase superscript on variable $X^{(Z)}$ additionally associates it with a specific relay mode Z . The subscript ψ denotes the fault class (e.g., primary fault).

The proposed method consists of the steps shown in Figure 3, which aim to address the goals outlined in Section 4. The relay setting groups are computed via offline steps prior to deployment. The offline steps consist of running Monte Carlo simulations to generate fault data representing the various fault scenarios that a protection relay may encounter following deployment. The fault data consist of local current and voltage measurements that are locally accessible to the relay and discrete system states, including switch and generation states, that can be communicated to the relay from the central MPMC when needed. Next, we apply the dimensionality reduction utility of GDA to identify the minimal communication requirements of the relay: we fit an LDA classifier to the fault data and extract the features of most importance to fault classification. We use these features to specify the grid state (or topology) changes that the MPMC needs to communicate to the relay to enable adaptive protection. A stage-2 LDA parametrizes the fault data to statistics that can be used in an optimization formulation to compute relay settings for relay selectivity and coordination. The setting groups are stored in the memory of each relay. Depending on the system

state, the relay operates in a certain mode with an associated setting group. Post-deployment, the MPMC communicates with the relay to switch its setting group only if the relay needs to adapt in response to a system change.

In the case of a fault, the relay computes its operation time based on local measurements and its relay setting group. The steps are explained in detail in the following subsections. We focus the section on the protection against three-phase faults as the strategy can be easily generalized to the protection against other types of faults. We discuss applicability to different fault types in Section 6.

5.1 Offline Steps

This subsection details the steps that are performed using offline simulation studies before deployment.

Generate Fault Data. Monte Carlo simulations [19] are performed to simulate various fault scenarios by leveraging probability distributions over factors that affect measured fault currents and voltages, including grid topology and generator availability and control, and uncertain factors such as fault impedance, fault distance from the relay, renewable energy intermittency, and fault class. Power system studies, historical data, or experts' subjective judgments should define these distributions.

A fault datum for relay r , of the form $\{\psi, I_F, V_F, \mathcal{X}\}^{(r)}$ consists of the fault current, I_F , and voltage, V_F , measured by the relay; and the fault class, ψ . In this work, we consider three classes for ψ grouped in the set $\Psi = \{\text{primary, backup, other}\}$; but these classes can be extended.

$\mathcal{X}$ is a vector of binary states depicting the grid configuration, including switch states (closed/open) and generator modes (in-service/out-of-service), which can be infrequently communicated to relay r . Each entry in $\mathcal{X}$ represents a distinct feature corresponding to the status of an equipment. Hence, each permutation of $\mathcal{X}$ represents a specific grid configuration; there are $2^{|\mathcal{X}|}$ total possible grid configurations that can be represented by $\mathcal{X}$, where $|\mathcal{X}|$ denotes the dimension of $\mathcal{X}$.

Specify Relay Communication. In the proposed method, each relay is an agent making decisions based on its local current and voltage measurement and current grid topology communicated to the relay from the MPMC. It is desirable to reduce the reliance (in terms of communication frequency and bandwidth) on communication infrastructure for protection decisions. To address this communication requirement, we use the feature importance, as determined by LDA, to specify which grid changes need to be communicated to the relay to adapt its settings. A (stage-1) LDA classifier fits Gaussian distributions to each class of ψ over the space $(\mathcal{X}, I_F, V_F)$, i.e.

$$(I_F, V_F) | \psi \sim \mathcal{N}(\mu_\psi^{(r)}, \Sigma^{(r)}), \quad \psi \in \Psi, \quad (8)$$

after which the features of $\mathcal{X}$ with the most significant coefficients in $\left\{ \left(\Sigma^{(r)} \right)^{-1} \mu_\psi^{(r)} \right\}^{(\psi \in \Psi)}$ are considered.

We denote the vector with reduced features $\mathcal{X}_S^{(r)}$, where $|\mathcal{X}_S^{(r)}| \leq |\mathcal{X}|$.

Each of the elements in $\mathcal{X}_S^{(r)}$ is an equipment status, which when changed requires the relay to use a different relay setting group. Hence, $\mathcal{X}_S^{(r)}$ contains information of the minimal distinct modes of operation of the relay. Grid changes that do not affect $\mathcal{X}_S^{(r)}$ and are not represented in its features would not significantly affect the protection decisions of relay r , and hence, are not communicated to the relay.

Note that each $(\Sigma^{(r)})^{-1} \mu_\psi^{(r)}$ yields a weights vector of size $|\mathcal{X}|$ representing the features' coefficients that define LDA's linear decision boundaries between the classes. These coefficients indicate the importance of each feature to the classification decision. To identify the importance order of the

features, we compute the mean coefficient for each feature across the fault classes $\psi \in \Psi$ and order the features according to the absolute values of their mean coefficients. We normalize the mean coefficients by dividing them by the maximum mean coefficient to compute the relative importance of the features. In this scale, the most important feature is assigned an importance value of 1. We select the most important features, beyond which there is no appreciable increase in the classifier's accuracy in predicting a fault class, to include in $\mathcal{X}_S^{(r)}$, discarding all other features. This process is illustrated in Section 6.1.

Subsequently, relay r will have $2^{|\mathcal{X}_S^{(r)}|}$ modes of operation, each with a unique group of relay settings, representing grid configurations that the relay needs to adjust to while neglecting other grid configurations that the relay need not adjust to. The relay modes will correspond to the permutations of the statuses of equipment represented by the selected important features in $|\mathcal{X}_S^{(r)}|$.

Parametrize Fault Data. Next, to parametrize the fault data into statistics that we can use for relay optimization and coordination, a (stage-2) LDA fits Gaussian distributions to each class of ψ over (I_F, V_F) for each relay mode of operation. For each relay mode Z , the distributions fitted to the data are

$$P(I_F, V_F | \psi, Z) = \mathcal{N}\left(I_F, V_F; \boldsymbol{\mu}_\psi^{(Z)}, \boldsymbol{\Sigma}_\psi^{(Z)}\right), \quad (9)$$

$$\psi \in \Psi, Z \in \mathcal{X}_S^{(r)}$$

where

$$\boldsymbol{\mu}_\psi^{(Z)} = \left[\mu\{I_F\}_\psi^{(Z)}, \mu\{V_F\}_\psi^{(Z)} \right]$$

$$\boldsymbol{\Sigma}_\psi^{(Z)} = \begin{bmatrix} \sigma^2\{I_F\}_\psi^{(Z)} & \sigma\{I_F, V_F\}_\psi^{(Z)} \\ \sigma\{I_F, V_F\}_\psi^{(Z)} & \sigma^2\{V_F\}_\psi^{(Z)} \end{bmatrix}$$

denote the mode's fault statistics: $\mu\{I_F\}_\psi^{(Z)}$ and $\mu\{V_F\}_\psi^{(Z)}$ are the means of the fault current and voltage, respectively, and $\sigma^2\{I_F\}_\psi^{(Z)}$ and $\sigma^2\{V_F\}_\psi^{(Z)}$ are the variances of the fault current and voltage, respectively, in mode Z .

Compute Relay Settings. Next, relay settings are optimized, using the above fault statistics, to achieve speed and selectivity of protection, and facilitate relay coordination. We describe the parameters of the optimization problem before formulating it. To simplify relay optimization, we assume $L_i = L_v = 0$ in Equation (1) and introduce the variable $\zeta = \text{TMS} \times \lambda_i \lambda_v$ to derive the following equation for relay operation time:

$$t_{op}(I_M, V_M; \zeta, \eta_i, \eta_v, I_S) = \frac{\zeta}{\left(\left(\frac{I_M}{I_S}\right)^{\eta_i} - 1\right) \left(\left(\frac{V_S}{V_M}\right)^{\eta_v} - 1\right)}. \quad (10)$$

The objective is to optimize the parameters ζ , η_i , and η_v .

To include information about both the mean and variance of the current and voltage fault data in the optimization problem, we replace I_M and V_M in Equation (10) with

$$I_M = \text{CVaR}_{\alpha_i}(I_F) = \mu\{I_F\}_\psi^{(Z)} - \sigma\{I_F\}_\psi^{(Z)} \frac{\phi(\Phi^{-1}(\alpha))}{\alpha} \quad (11)$$

$$V_M = \text{CVaR}_{\alpha_v}(V_F) = \mu\{V_F\}_\psi^{(Z)} - \sigma\{V_F\}_\psi^{(Z)} \frac{\phi(\Phi^{-1}(\alpha))}{\alpha}. \quad (12)$$

Using the CVaR enables more control over the parametrization of the fault statistics, as we illustrate in Section 6.

We use the following notation in the optimization problem formulation: $\mathcal{R}$ is the set of all relays in the system, $\mathcal{B}$ is the set of all buses in the system, $\mathcal{P}_B \subset \mathcal{R}$ is the set of all primary relays to bus B , and $\mathcal{S}_p \subset \mathcal{R}$ is the set of all backup relays to relay p . ζ is the vector of all settings $\zeta^{(r)}$, $r \in \mathcal{R}$. η_i, η_v are the same for all relays in the system for a specified topology.

Primary relays should operate following a short delay $D_{\min}$, and backup relays should operate following an MCT delay.

For all grid topology configurations, we solve the following:

$$\min_{\zeta, \eta} \sum_{B \in \mathcal{B}} \left(\sum_{p \in \mathcal{P}_B} \left(t_{op}(\text{CVaR}_{\alpha_i}^{(p)}, \text{CVaR}_{\alpha_v}^{(p)}; \zeta^{(p)}) + \sum_{b \in \mathcal{S}_p} t_{op}(\text{CVaR}_{\alpha_i}^{(b)}, \text{CVaR}_{\alpha_v}^{(b)}; \zeta^{(b)}) \right) \right) \quad (13)$$

subject to

$$\eta_{i,\min} \leq \eta_i \leq \eta_{i,\max} \quad (14)$$

$$\eta_{v,\min} \leq \eta_v \leq \eta_{v,\max} \quad (15)$$

$$\zeta_{\min} \leq \zeta^{(r)} \leq \zeta_{\max}, \quad r \in \mathcal{R} \quad (16)$$

$$t_{op}(\text{CVaR}_{\alpha_i}; \zeta^{(p)}) \geq D_{\min} \quad (17)$$

$$t_{op}(\text{CVaR}_{\alpha_2}; \zeta^{(b)}) - t_{op}(\text{CVaR}_{\alpha_i}; \zeta^{(p)}) \geq \text{MCT}, \quad (18)$$

$$B \in \mathcal{B}, p \in \mathcal{R}_B, b \in \mathcal{S}_p$$

Next, relay settings $\{\zeta^{(r)}, \eta_i, \eta_v\}$ are stored in-memory on relay r if the grid topology is expressed in the relay's modes of operation.

This optimization problem over $\{\zeta, \eta_i, \eta_v\}$ is not convex. However, pre-setting η_i, η_v yields a linear program over ζ . We optimize η_i, η_v via gradient descent by re-running the linear program for different values of η_i, η_v . While not convex, in practice, we have found that starting from very small $\eta_i = \eta_v = 0.01$ yields good optimization results.

The pickup settings I_S, V_S are preset before optimization. I_S should exceed the maximum full-load line currents, and V_S should be less than bus voltages seen during normal operation. We compute I_S separately for each relay depending on its line's maximum load, and preset V_S to 0.9 pu for all relays.

5.2 Online Operation

During operation, each relay operates in a specific relay mode with a relay setting group, based on the current state of the system. If the system state changes and the relay's fault response needs to be switched accordingly, the MPMC communicates with the relay to instruct it to switch to a different relay mode. The communication between the MPMC and the relay should be developed to ensure that the MPMC message has been delivered successfully to the relay (e.g., acknowledgments); otherwise, the message should be redelivered.

In the event of a fault, the relay detects the fault by checking whether its local current and voltage drop measurements exceed its pickup settings. Applying Equation (10), the relay calculates its operation time. If the fault is not cleared before the relay's operation time is reached, the relay trips its circuit breaker. If the relay fails to trip, backup relays compute a longer operation time and trip their circuit breakers following an MCT delay. After tripping its circuit breaker, the relay communicates to the MPMC the open status of its circuit breaker.

6 Simulation Results, Validation, and Discussion

In this section, we illustrate the proposed method and discuss its benefits.

As discussed in Section 4, the benchmark system we use to demonstrate the results is the CIGRE medium voltage distribution network. The system is shown in Figure 2. The system-rated voltage and frequency are 12.47 kV and 60 Hz. To study fault current contributions from distributed generation, we add, as in [20], three DG units:

- (1) DG1 is a 4 MW rated full-scale Voltage-source Converter-interfaced DG. A 5 MVA, 4.16/12.47 kV, $X = 0.07$ pu dYG transformer connects the DG to bus 6.
- (2) DG2 is composed of two 4.3 MVA synchronous generators. A 10 MVA, 4.16/12.47 kV, $X = 0.07$ pu dYG transformer connects the DG to bus 8.
- (3) DG3 is a 4.3 MVA synchronous generator. A 5 MVA, 4.16/12.47 kV, $X = 0.07$ pu dYG transformer connects the DG to bus 12.

We generate short circuit data using Pandapower [45]. Pandapower is an open-source software library for power system analysis and optimization in Python, which we use for the ease of integration with Python's data analysis and ML libraries. Pandapower's short circuit calculations are based on IEC60909 [22] and tested against DigSilent. We also verify our results using a PSCAD time-domain fault simulation. The probability distributions used to generate Monte Carlo simulations for faults with different grid topology and fault impedance are outlined in Table 2 in Appendix.

The studied cases focus on relay R12-13 and the protection of bus 8 for the insights that can be gained from their analysis. The study of R12-13 validates the method for adjusting relay settings in response to variations in fault current resulting from transitioning from an isolated to a grid-connected mode of operation, as well as modifications in network structure from radial to mesh configuration. The study of bus 8 faults validates the method for coordinating relays in a mesh topology. In optimizing the relay settings, we set $D_{\min} = 3$ cycles, $MCT = 0.15$ seconds, and $\alpha_i = \alpha_o = 1$.

6.1 Computing Offline Relay Settings of R12-13

Figure 4 illustrates **Kernel Density Estimates (KDEs)** of the fault current and voltage data of relay R12-13. The independent variables in Figure 4(a) and (b) represent the fault currents and voltages measured by R12-13 during primary, secondary, and other classes of faults. Figure 4(c) combines the currents and voltages into a single KDE plot. Note that the current (voltage) is expressed in terms of multiples (inverse multiples) of the current (voltage) pickup setting. The significant overlap between the different fault classes hinders the ability to classify faults based on the fault current and/or voltage magnitude. To address this issue, stage-1 LDA separates the relay operation into modes that better distinguish the fault classes based on current and/or voltage measurement to enable a local measurements-based approach for protection.

Figure 5 displays the feature importance coefficients as determined by the stage-1 LDA. From Figure 5, it is observed that the features with the most significant importance coefficients are the voltage and current magnitudes, and the status of the downstream switch S1. This is further confirmed in Figure 6 by evaluating the LDA's classification accuracy as the number of features it has access to is increased (in accordance with the order of feature importance in Figure 5).

Each column in Figure 6 represents the classification accuracy of the stage-1 LDA model when trained on the feature specified under that column and the features under all columns to its left. For instance, the fourth column, labeled PC2, indicates the classification accuracy when trained on data solely described by voltage multiples (V), current multiples (I), and the states of S1 and PC2.

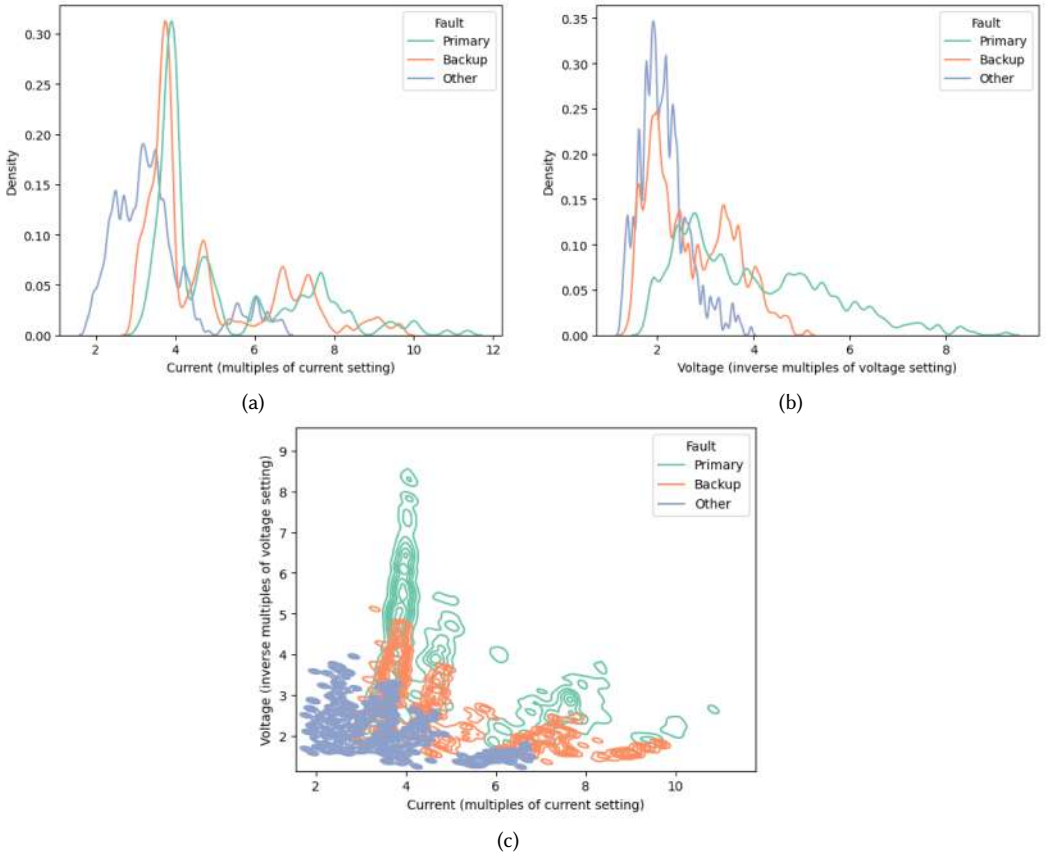


Fig. 4. KDE plots of (a) current multiples, (b) voltage inverse multiples, and (c) current multiples and voltage inverse multiples measured by relay R12-13 during three-phase faults.

The order of features in Figure 6 (from left to right) corresponds to the importance of the features identified by the stage-1 LDA, as shown in Figure 5 (from bottom to top).

The blue bars in Figure 6 indicate the accuracy when classifying faults into three classes (primary, secondary, or other), and the orange bars indicate the accuracy when classifying faults into two classes (primary/secondary or other). The classifier's accuracy (in blue) reaches a plateau of 77% after being provided with a small subset of the signals comprising the voltage and current magnitudes, and the states of S1 and PC2.

This provides evidence that the other features do not contribute any appreciable performance gains and can be disregarded to reduce memory and communication requirements with relay R12-13.

As such, we compartmentalize R12-13's operation into $2^2 = 4$ modes of operation corresponding to permutations of each of S1 and PC2 being open or closed. Hence, the MPMC needs to only communicate the statuses of S1 and PC2 to relay R12-13.

In Figure 7(a)–(d), we divide R12-13's operation into these 4 modes and plot the KDEs of the fault current and voltage data in each mode. Figure 7(a) and (b) correspond to the modes with S1 open; hence, R12-13 only measures primary and secondary faults. Figure 7(b) and (d) correspond to

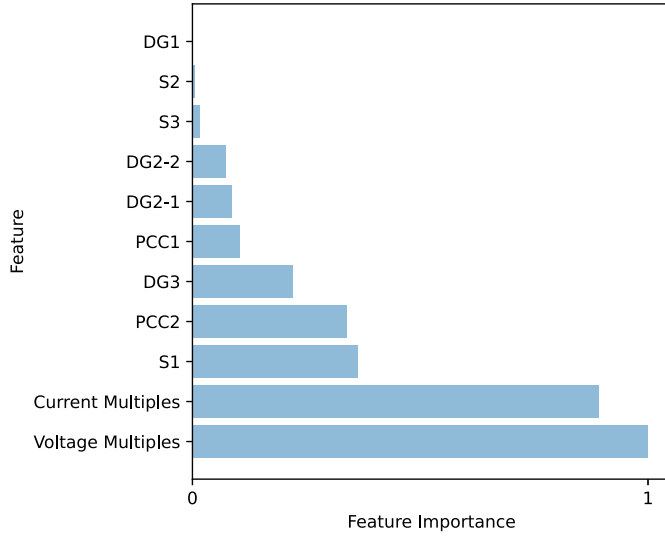


Fig. 5. Feature importance coefficients of the GDA classifier trained on R12-13 fault data. The voltage and current are the only two local measurements. The rest are states of switches (S), points-of-common coupling (PC), or DGs that can be communicated to the relay from the MPMC. The horizontal axis depicts the importance relative to the most important feature.

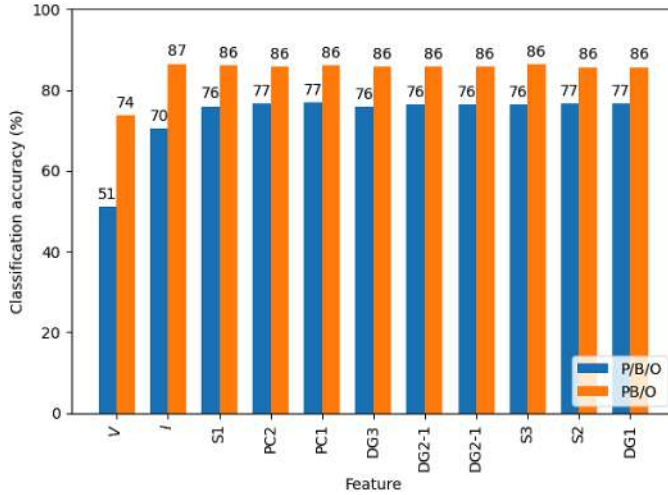


Fig. 6. GDA fault classification accuracy. The classes are primary (P), backup (B), and other (O). Each column represents the classification accuracy when the GDA is trained on its feature, in addition to all features to the left of the column. The right-most column is the GDA accuracy when trained on all features. The accuracy plateaus after the first four features.

the modes with PC2 closed; hence, R12-13 experiences higher fault current magnitudes due to the increased contribution from upstream sources, including the main grid.

It can be observed that by compartmentalizing the relay operation into modes, the overlap between the different fault classes is significantly reduced, facilitating the distinction of fault classes based on voltage and current magnitude. The fault current (voltage) magnitude decreases (increases)

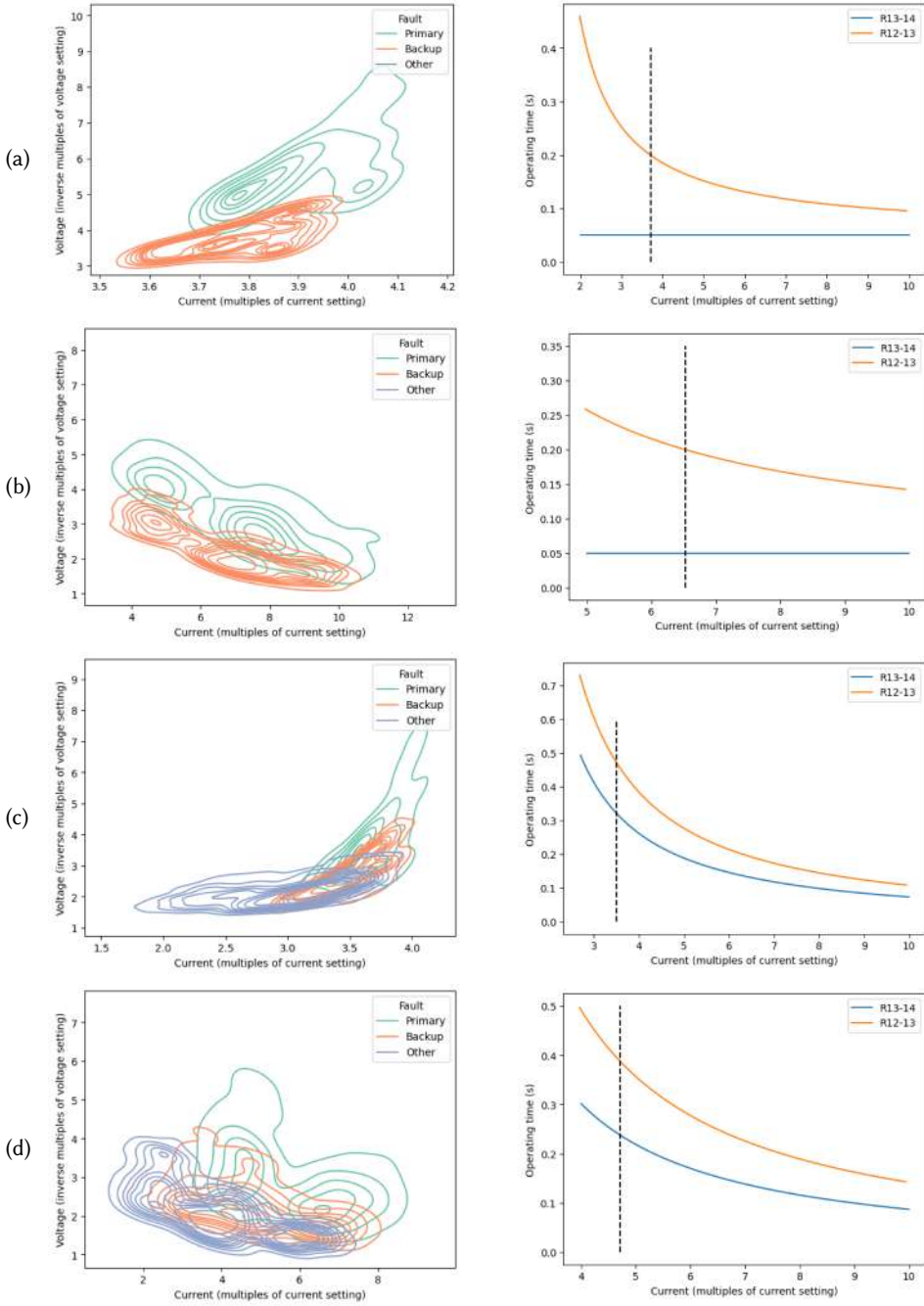


Fig. 7. (left) KDE plots of current multiples and voltage inverse multiples measured by relay R12-13 and (right) relay operating curves against current multiples for each of its four modes. Modes: (a) S1, PC2 open, (b) S1 open, PC2 closed, (c) S1 closed, PC2 open, and (d) S1, PC2 closed.

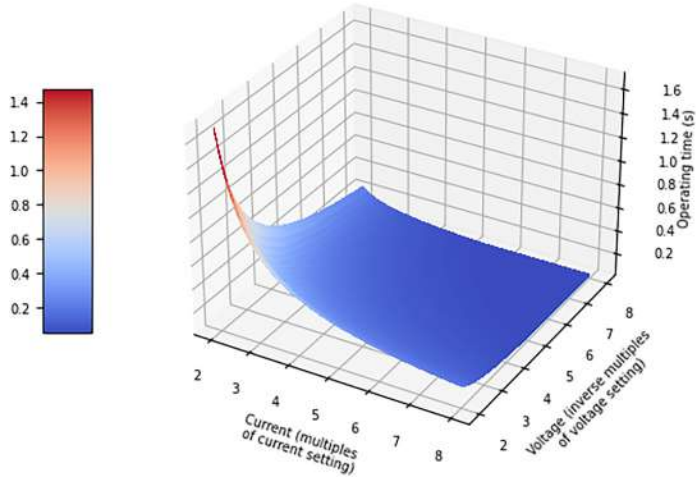


Fig. 8. Relay operating curve of relay R12-13 against current multiples and voltage inverse multiples for mode (c): S1 closed, PC2 open.

as the distance from the fault increases. This enables the use of inverse-time protection to coordinate relays. The objective of the stage-2 LDA is to parametrize the fault data so that their distribution statistics can be utilized for relay coordination.

To the right of each KDE plot in Figure 7, we plot the corresponding inverse-time operation curves against the fault current (measured as multiples of pickup current setting). While the relay operation time is a function of both the fault current and voltage, we only plot the operation curves against the fault current to demonstrate relay coordination, as it is hard to observe the coordination in a three-dimensional plot. We illustrate the full three-dimensional inverse-time operation curve of R12-13 against both fault current and voltage (measured as inverse multiples of pickup voltage setting) for mode (c) in Figure 8.

To demonstrate relay coordination:

- We plot vertical dashed lines in Figure 7 to illustrate the CVaR of primary fault currents measured by R13-14 (blue), i.e., $\text{CVaR}_{\alpha_I}(I_F)^{13-14}$.
- Since the current multiple measurements by the backup relay R12-13 might be different than that measured by R13-14 due to different I_S settings or line branching, we shift the operation curve of R12-13 so that the vertical line also matches how R12-13 measures the fault current.
- The voltage value of each of the operation curves of R13-14 and R12-13 corresponds to the voltage values each measure during the fault characterized by the vertical line.

The intersection of the relay curve with the dashed line marks the relay operation time; hence, the higher curve trips later. Relay coordination can be easily evaluated by looking at the relay operation curves. R12-13 serves as a backup for R13-14, which can be observed from R12-13's curve being higher than R13-14's curve in all modes.

Figure 7 illustrates how the relays adjust to changes in the grid operation, including going from connected to islanded modes and switching between radial and meshed topologies. The four relay group settings for R12-13 and its downstream relay R13-14 are included in Table 3 in Appendix. These settings groups must be stored in their memories to enable them to adapt to changing conditions.

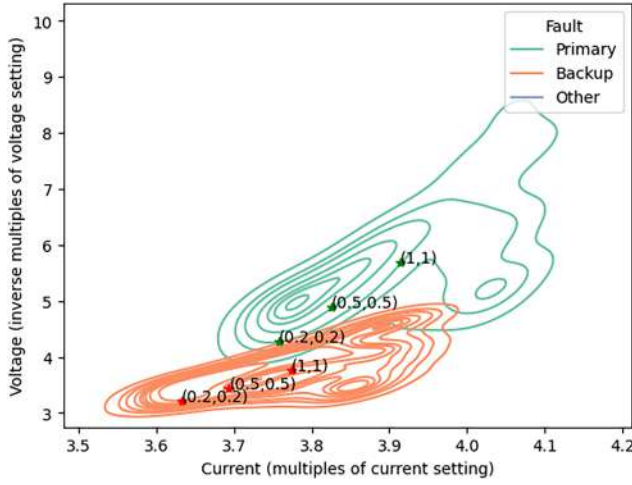


Fig. 9. Illustrating the benefits of CVaR in optimization. Each point represents $(\text{CVaR}_{\alpha_i}(I_F))^{12-13}$, $\text{CVaR}_{\alpha_v}(V_F)^{12-13}$. The values of (α_i, α_v) are annotated.

Remarks: In modes (a) and (b) when S1 is open, the optimization results in a TMS equal to 0 for R13-14, which means it is sufficient to operate R13-14 as an instantaneous over-current relay. The increase in operation time for the relays when S1 is closed allows for coordination with relays further downstream.

In Figure 7(c), an important insight is the distinctive “crescent” shape observed in the KDE plot. This shape indicates that voltage drops increase more significantly than fault current increases for nearer faults. A factor contributing to this is the limited fault current of DGs. As a result, incorporating voltage measurements enhances the ability to distinguish fault classes in this mode. This finding supports our decision to include voltage drops in the inverse-time relay operation.

Lastly, we discuss how the use of CVaR is beneficial in the relay optimization process. Figure 9 shows how different value pairs of CVaR for current and voltage can be used to better control how the data are parametrized before it is used in the optimization. Lowering the CVaR values allows for greater emphasis on the lower tails of the fault current and voltage distributions. This is useful because inverse-time protection is faster at clearing faults with high current and low voltage magnitudes (top-right corner of the figure). Lowering (raising) the CVaR values shifts the focus of the optimization and allows for faster (slower) clearing of faults.

6.2 Relay Coordination at Bus 8 with Topology Change

To validate the method for coordinating relays in a mesh topology, we consider faults at bus 8. When S1 and S2 are closed, a fault at bus 8 is isolated by three primary relays: R14-8, R7-8, and R3-8. Each primary relay has one or more backups.

In Figure 10, we plot the relay operation curves of relays (a) R14-8, (b) R7-8, and (c) R3-8 and their backup relays when PC1, PC2, S1, and S2 are closed. The dashed vertical lines show the CVaR of fault currents measured by the primary relays for faults at bus 8. The figure demonstrates the successful isolation of the fault from all short-circuit current sources and relay coordination in case of relay failure.

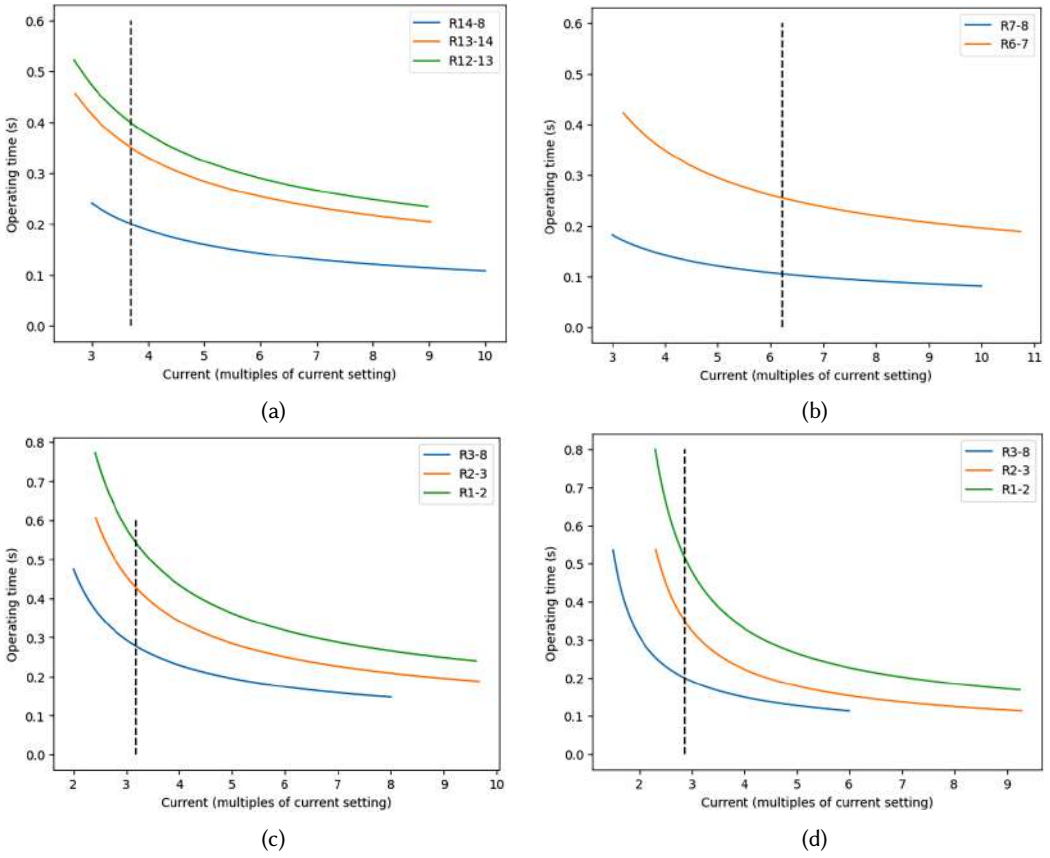


Fig. 10. Relay operation curves against current multiples for relays (a) R14-8, R13-14, and R12-13; (b) R7-8 and R6-7; and (c) R3-8, R2-3, and R1-2 for a fault at bus 8 when S3 is open and S1, S2, PC1, and PC2 are closed. (d) Relay operation curves against current multiples for relays R3-8, R2-3, and R1-2 for a fault at bus 8 when S1, S2, and S3 are open and PC1 is closed.

When S1 and S2 are open, the topology of the microgrids becomes radial with R3-8 offering primary protection to bus 8 and R2-3 and R1-2 its backups. The fault current is supplied by the main grid. The relay operation curves of these relays in this case are shown in Figure 10(d). PSCAD simulations validate the coordination results of this case study. In Figure 11, a 1Ω fault occurs at bus 8 at the 0.2 seconds mark. We suppress the operation of the primary (R3-8) and its backup (R2-3) relays, and let R1-2 isolate the fault. The operation times of the relays closely match the coordination curves in Figure 10(d).

The relay settings for these two case studies are listed in Table 4 in Appendix.

6.3 Discussion

The proposed method has the following merits:

6.3.1 Evaluating LDA Accuracy. The significant overlap of different fault classes observed in Figure 4 shows that it is infeasible to classify a fault based only on local current and voltage measurements. In Figure 12, we compare the classification accuracy of multiple ML algorithms in classifying the faults measured by R12-13 when limiting the training data to discrete grid states and local measurements—to avoid the reliance on real-time communication. On the left side of Figure 12,

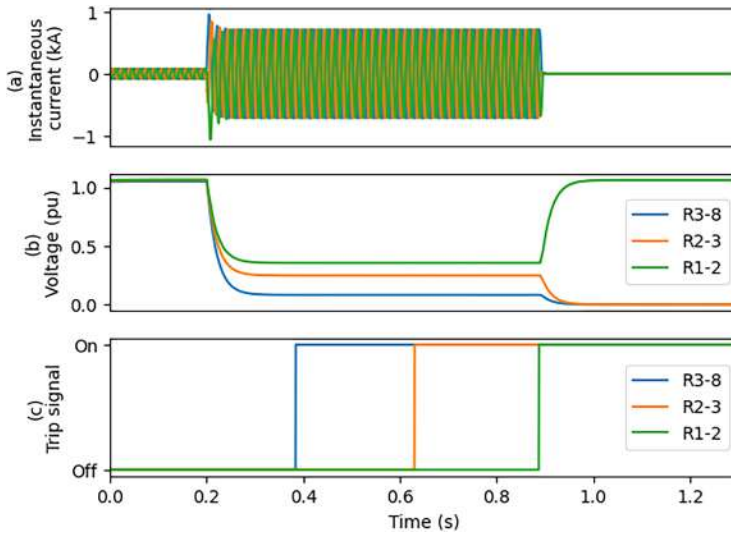


Fig. 11. PSCAD simulation of the (a) instantaneous current measured by R1-2, (b) voltages, and (c) tripping signals of R3-8, R2-3, and R1-2 for a 1Ω fault at bus 8 with PC1 closed, and switches S1, S2, and S3 open.

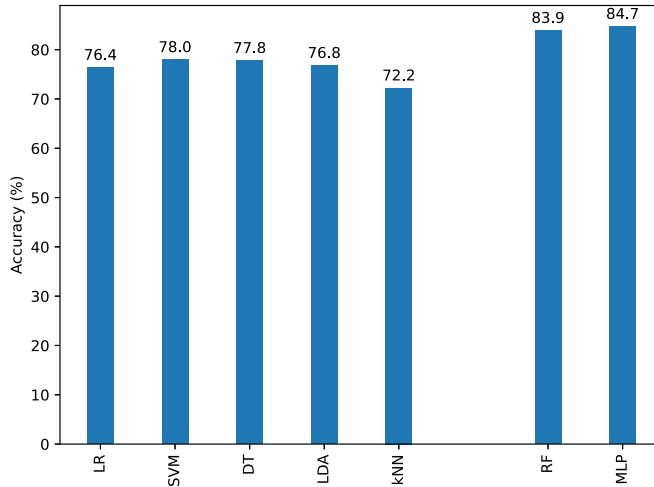


Fig. 12. Comparing LDA classification accuracy with other algorithms.

we include algorithms that yield interpretable results, including **Logistic Regression (LR)**, SVM, DT, LDA, and kNN. On the right, we include algorithms that generally have lower learning bias and higher accuracy but do not yield interpretable results, including RFs and **Multi-layer Perceptrons (MLPs)**. Hyperparameter optimization for these algorithms is listed in Table 5 in Appendix. We observe that MLPs achieve the highest classification accuracy, as expected, but highlighting the difficulty of the classification task, the maximum accuracy is only 84.7%.

Note that these algorithms have been shown to yield higher performance in centralized real-time communication-based protection strategies in the literature when not constrained to local measurements and discrete system states. However, this exposes the protection to reliability threats related to the communication infrastructure.

While LDA's accuracy is lower, the interpretability of LDA's results enabled us to augment it with an optimization-based approach that facilitates relay coordination, preventing miscoordination between relays and their backups. The interpretability of LDA enabled easy visualization of the fault data statistics, protection decisions, and relay coordination. Hence, the results of the method can be easily interpreted and validated by system operators and protection engineers, facilitating its use for adaptive protection and reducing associated implementation risks and reliability threats.

Despite its lower accuracy, we prefer LDA over DTs due to its stability. Further, we prefer LDA over SVM since LDA enables fitting Gaussian distributions to fault classes, which we use to reduce problem dimensionality and parametrize the fault data for optimization. As observed by the results, Gaussian distributions fit well to the fault data.

6.3.2 Feature Selection. As an alternative to employing LDA for selecting the features that are most important to determining the relay's mode of operation—for specifying communication requirements, previous literature has explored feature selection techniques, such as RFE and meta-transformers [34, 38, 42].

Both RFE and mTs leverage the weights assigned by a classification model to features. RFE assesses feature importance by recursively considering smaller sets of features, iteratively eliminating the least important features until reaching the desired number of selected features. In the process, RFE produces an importance ranking of the features. On the other hand, mTs use weights to gauge importance and eliminate features with importance below a specified threshold parameter. Typically, this threshold corresponds to the mean importance of all features.

In Table 1, we present a comparison between RFE and a mT's results for feature selection using the following models: LDA, LR, SVM, and RFs—commonly employed in adaptive protection research. To facilitate demonstration and comparison with Figure 5 results, we specifically showcase the outcomes for relay R12-13.

Table 1 reveals that RFE consistently ranks local current and voltage measurements, along with the states of S1 and PC2, as the most important features, supporting our results. mTs align with this selection when implementing LR and LDA, while the SVM mT opts for three features, prioritizing the state of PC2 over S1. Figure 13(a) and (b) illustrate that with these four features, LR and SVM achieve classification accuracies of 75.6% and 77%, respectively—comparable to LDA's 77% accuracy presented in Figure 6.

Although RFs lack interpretability, they still facilitate an understanding of feature importance. The important features are those that minimize the entropy of the trees' leaves. The RF mT identifies local current and voltage measurements as the most important features, with which the RF classifier attains an accuracy of 80.4% as demonstrated in Figure 13(c).

In conclusion, our approach to specifying communication requirements and relay modes of operation is comparable to methods proposed in existing literature.

6.3.3 Communication Network Delays and Cyberattacks. The proposed protection strategy requires communication of the grid topology to the relays. Generally, this reliance on communication infrastructure can expose the protection relay operation to potential risks, such as network delays or cyberattacks. To address this issue, our method minimizes communication to the relays by communicating only a subset of the infrequent major changes to the grid topology. The necessary settings to operate the relays in real-time are stored locally in memory on the relays. This provides an advantage over centralized protection strategies that require a constantly active communication channel with the relay. Due to the infrequent communication, temporary network delays do not significantly impact the protection. Further, the infrequency permits the use of encryption algorithms to authenticate and protect the communication to relays. This approach ensures that

Table 1. Feature Importance Order in the Order of Most Important to Least Important from Top to Bottom

LDA		LR		SVM		RF	
<i>RFE</i>	<i>mT</i>	<i>RFE</i>	<i>mT</i>	<i>RFE</i>	<i>mT</i>	<i>RFE</i>	<i>mT</i>
<i>V</i>	<i>V</i>	<i>V</i>	<i>V</i>	<i>V</i>	<i>V</i>	<i>I</i>	<i>I</i>
<i>I</i>	<i>I</i>	<i>I</i>	<i>V</i>	<i>I</i>	<i>I</i>	<i>V</i>	<i>V</i>
S1	S1	S1	S1	S1		S1	
PC2	PC2	PC2	PC2	PC2	PC2	PC2	
DG3		DG3		DG3		PC1	
PC1		PC1		PC1		DG3	
DG2		DG2		DG2		DG2	
S3		S3		S2		S3	
S2		S2		S3		S2	
DG1		DG1		DG1		DG1	

Each classification model has two columns under it corresponding to the feature importance ranking per RFE and important feature trimming per mT.

the proposed protection strategy is more secure against potential cyberattacks and/or network delays.

6.3.4 Achieving Protection Relay Characteristics. The proposed protection strategy extends the operation of inverse-time protection relays by adapting protection decisions to changing grid conditions and considering voltages for an improved distinction of faults in systems with limited fault currents. Below, we evaluate the proposed strategy against the characteristics of effective power system protection strategies.

- Reliability: Within the context of protection systems, reliability mandates the ability to operate effectively during the presence of faults while abstaining from unnecessary operation in the absence of faults. Our devised protection strategy prioritizes verifiable reliability through several measures.

Firstly, the utilization of the inverse-time equation (10) in determining relay operation ensures that relays are triggered only when the fault current exceeds a predefined pick-up current setting and the voltage drops below a predefined pick-up voltage setting. This minimizes the risk of false operation in the absence of faults.

Secondly, our proposed strategy (specifically, the optimization (13)) yields optimal relay settings that generate easily interpretable relay operating curves, exemplified in Figures 7 and 10, facilitating the review and validation of the protection decisions. The optimized settings guarantee a consistently rapid response to faults, as evident in Figures 7 and 10. This rapid response is further supported by the time-domain simulation in Figure 11, which shows the primary relay responding to the fault in less than 200 milliseconds and three levels of sequential protection responding to the fault in less than 800 milliseconds.

Moreover, our proposed protection strategy allows high flexibility in accommodating diverse considerations for fault response to enhance protection reliability. Monte Carlo simulations can model uncertain variables like fault impedance, inception angles, pre-fault power system conditions, and relay measurement errors. Additional considerations, such as in-rush currents

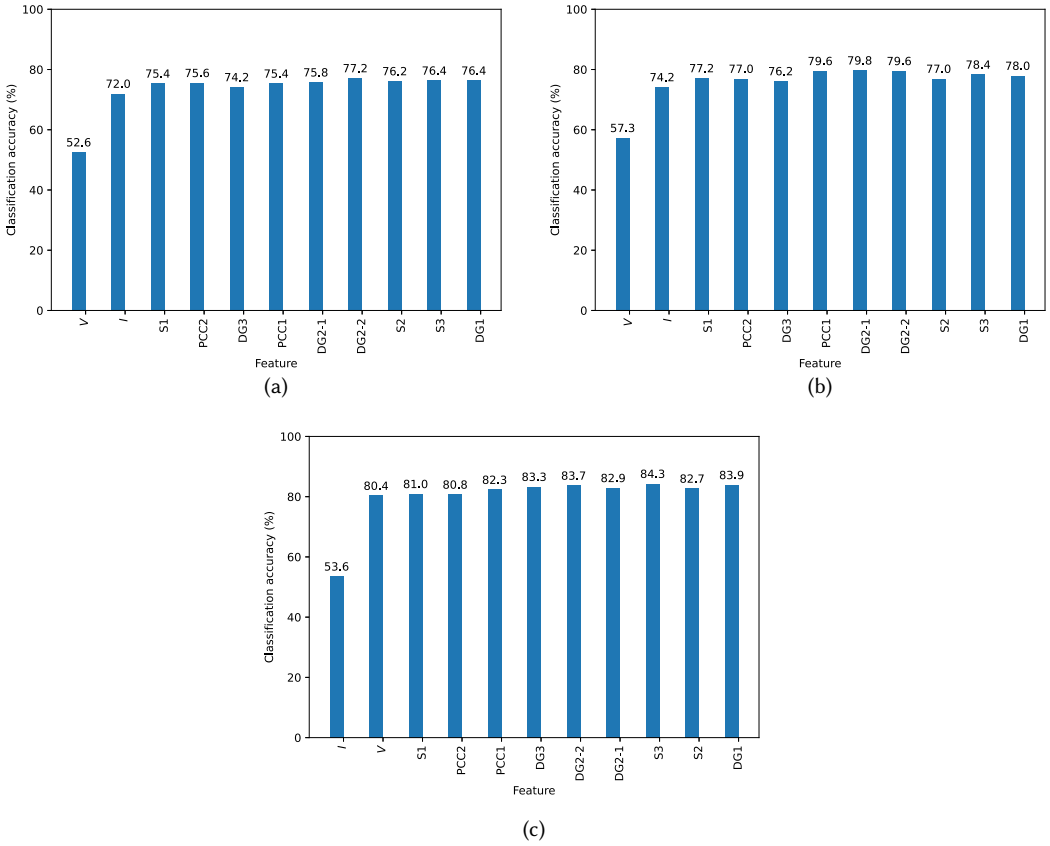


Fig. 13. Fault classification accuracy with select features for (a) LR, (b) SVM, and (c) RF models. Each column represents the classification accuracy when the model is trained on its feature, in addition to all features to the left of the column. The right-most columns depict the accuracy when trained on all features. The order of features on the horizontal axes matches the order of importance of features in Table 1.

from load switching, distributed generation, filter capacitors, or transformers, along with transformer saturation, can be addressed similarly to conventional over-current relays or incorporated as constraints in the optimization (13). The inclusion of the CVaR approach permits fine-tuning of relay operation times as showcased in Figure 9.

Lastly, our method facilitates the visualization and comprehension of (worst-case) longest potential fault-clearing times. The tails of the distributions in the bottom left corners of the KDE plots represent faults with the longest clearing times. This visualization aids in understanding the operation time ranges of the protection relays.

- **Selectivity and Coordination:** Selectivity aims to minimally isolate the faulty part of the system, while coordination ensures a sequential and well-timed activation of protective devices.

Our proposed strategy facilitates the complex coordination of relays. The resulting selectivity and coordination of the protection strategy is exemplified in Figure 10, where the isolation of a fault in a mesh topology is achieved through the coordinated tripping of three different primary relays. Additionally, five backup relays are prepared to operate in case of primary relay malfunction, showcasing a coordinated and layered protection system.

- **Adaptability:** The relay curves depicted in Figure 7 demonstrate the adaptability of the protection strategy. Following transition from islanded to grid-connected mode, the protection relays adapt to the changing short-circuit current capacity to ensure consistently rapid response times to faults in both grid conditions. Furthermore, when transitioning from a radial to a mesh topology, the protection devices exhibit adaptability by coordinating with other relays within the mesh structure. Notably, in a radial topology with S1 open, relay R13-14 operates without the need for coordination with any downstream relays. In this scenario, the protection strategy designates it as an instantaneous over-current relay with a flat relay curve. Conversely, in a mesh topology, R13-14 requires coordination with other relays. Accordingly, the protection strategy adapts the operating curves of R13-14 to facilitate effective coordination within the mesh structure. This flexibility in adjusting relay operating characteristics underscores the protection strategy's ability to adapt its response based on the specific grid topology.
- **Simplicity:** Modern **Intelligent Electronic Devices (IEDs)** used for protection relays can readily process the proposed protection strategy. The strategy minimizes the reliance on the communication infrastructure and memory use of the IEDs.

In addition, the method yields interpretable decisions that can be easily validated by protection engineers.

- **Communication:** The strategy relies on a communication infrastructure between the MPMC and relays. Commands can be established from the MPMC to switch the relay setting group by means of the IEC61850 protocol.

The method minimizes the communication bandwidth and frequency, only relying on infrequent updates due to changes in grid topology.

6.3.5 Applicability to Different Protection Relays and Fault Types. The proposed method specifies the communication requirements and optimizes relay settings. While we demonstrated our method on inverse-time protection relays, the method can be extended to other protection devices, such as distance relays, to adapt their operation to changing grid conditions. We emphasize that the main contribution of the article is the data-driven component: applying GDA to the adaptive protection of power systems.

We demonstrated the strategy for three-phase faults. The proposed strategy applies to other fault types. In the same way that relay settings were computed for three phase faults in the article, data can be generated and relay settings can be computed for other fault types. The relay settings are stored in the relay's memory. During a fault, the relay uses the current and voltage measurement to distinguish the fault type. The relay settings for the fault type and relay mode are used to compute an operation time and isolate the faulted phases.

7 Conclusion

In recent years, there has been a growing interest in utilizing ML for adaptive protection in power systems, leveraging the increasing amount of power system data. However, safety-critical applications such as power system protection require methods that are transparent, interpretable, and easily validated. Deep learning methods, although highly accurate, are difficult to validate—presenting a significant risk to their utilization for power system protection.

To address this issue, we propose a novel adaptive protection strategy for inverse-time relays based on probabilistic ML, specifically GDA. We apply the strategy to distill power system data to inverse-time relay settings, generating operation curves that can be used to verify and validate protection decisions. We demonstrate the use of our strategy for fault detection and localization, relay coordination and optimization, and protection management by specifying minimal communication requirements for adapting relays. To demonstrate the effectiveness of our method, we apply

it to the protection of the CIGRE medium voltage grid, as it switches between islanded and grid-connected modes of operation, as well as radial and mesh topology in the presence of distributed generation.

Appendix

Table 2. Monte Carlo Simulation Variations

Random variable	Values
Fault resistance	Unif(0, 3)Ω
Fault reactance	Unif(0, 3)Ω
Main grid short-circuit capacity	20–22 MVA
DG1 short circuit to nominal current ratio	1.1–1.3
Switch status	On/off
PC states	On/off
Generator in-service	True/false

Table 3. Relay Settings for Relays R12-13 and R13-14

Mode	η_i	η_v	ζ^{13-14}	ζ^{12-13}
S1 open, PC2 open	1	1	0	0.81
S1 open, PC2 closed	1	1	0	0.62
S1 closed, PC2 open	1.24	0.9	2.03	2.52
S1 closed, PC2 closed	1.2	1.01	3.17	2.27

Table 4. Settings of Relays Protecting Bus 8

Mode	η_i	η_v	Primary	Backup
Open: S1, S2, S3 Closed: PC1, PC2	0.1	2.89	$\zeta^{3-8} = 5.07$	$\zeta^{2-3} = 0.66$
Closed: S1, S2 PC1, PC2	0.1	4.75	$\zeta^{3-8} = 7.03$ $\zeta^{14-8} = 3.25$	$\zeta^{2-3} = 2.26$ $\zeta^{13-14} = 1.85$
Open: S3			$\zeta^{7-8} = 6.75$	$\zeta^{6-7} = 14.91$

Table 5. Hyperparameter Optimization

Algorithm	Hyperparameters	Range	Optimal
RF	Number of DTs	5–40	30
kNN	Number of neighbors	2–10	8
MLP	Hidden layers	5–80	40

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